

The CKM Wolfenstein Parameters from Generative Triple Evolution Orbit Arithmetic

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Abstract

We derive all four Wolfenstein parameters of the CKM quark mixing matrix from the arithmetic of the Rule 110 cellular automaton orbit that the Standard Model generation sequence is forced to satisfy [1], with zero free parameters. The derivation proceeds in two stages. First, the five quark-sector orbit indices N_{eff} are identified with structural formulas in the GTE orbit constants $N_{\text{gen}} = 3$, $N_{\text{fam}} = 5$, $c_H = 13$: the up quark at $b_u = N_{\text{gen}}^2 = 9$; the down quark at $b_d = N_{\text{fam}} = 5$; the charm quark at $b_c = N_{\text{fam}}^2(2N_{\text{fam}} + 1) = 275$; the strange quark at $b_s = 2N_{\text{gen}}(2c_H + N_{\text{fam}}) = 186$; and the bottom quark at $b_b = 2^{c_H} - 1 = 8191$, a Mersenne prime at the Higgs staircase endpoint. None of these values is a free parameter; all are consequences of the GTE cascade structure, machine-certified in Lean 4 with zero `sorry` for the arithmetic identities.

Second, the four Wolfenstein parameters follow. The leading parameter $\lambda = N_{\text{gen}}^2/(2^{N_{\text{gen}}} \times N_{\text{fam}}) = 9/40 = 0.22500$ agrees with the PDG value 0.22500 ± 0.00067 to **0.000%** (0.000σ); it is machine-certified in Lean 4 (`wolfenstein_lambda_formula`, zero `sorry`). The subleading parameter $A = \sqrt{N_{\text{eff}}(s)/N_{\text{eff}}(c)} = \sqrt{186/275} = 0.8224$ matches PDG at 0.65σ . The remaining parameters $\bar{\rho}$ and $\bar{\eta}$ follow from a deep cross-sector identity: the CKM unitarity triangle radius $R_b = N_{\text{gen}}/2^{N_{\text{gen}}} = 3/8$ equals the GUT-scale Weinberg mixing angle $\sin^2\theta_W(M_{\text{GUT}}) = 3/8$ (machine-certified; `ckm_unitarity_triangle_radius_eq_gut_weinberg`). From $R_b = 3/8$ and a Mersenne-to-algebraic ratio $\tan(\gamma) = \sqrt{b_b/b_s}/N_{\text{gen}}$, we obtain $\bar{\rho} = 0.1545$ (0.41σ) and $\bar{\eta} = 0.3417$ (0.63σ). All nine CKM matrix elements agree with the PDG within 1σ at $O(\lambda^4)$. The CP angle $\gamma = 65.67^\circ$ lies -0.023σ from the PDG central value. The Mersenne primality of b_b forces $\tan(\gamma)$ to be irrational, providing an arithmetic origin for CP violation.

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Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A}/\mathbf{D} > \mathbf{B} > \mathbf{D}$$

A_{Lean}: machine-checked exact arithmetic in Lean 4, zero **sorry**.

A_{MDL}: MDL-unique within a declared expression class.

A/D: computationally confirmed; a physics bridge remains.

B: calibrated reproduction.

D: exploratory / frontier.

Table 1: Reviewer’s map: principal results, claim strengths, and locations.

Result	Status	Claim	Where
$\lambda = 9/40 = 0.22500$	A_{Lean}	Lean-certified (<code>wolfenstein_lambda_formula</code> , zero sorry); 0.000σ from PDG.	§4
$A = \sqrt{186/275} = 0.8224$	A	Computationally verified; 0.65σ from PDG. Lean cert pending.	§5
$R_b = \frac{3}{8} = \frac{\sin^2\theta_W(M_{\text{GUT}})}{\sin^2\theta_W(M_{\text{GUT}})}$	A_{Lean}	Cross-sector identity; alias of <code>gut_weinberg_angle_pow2</code> via <code>ngen_plus_nfam_eq_pow2</code> , zero sorry .	§6
$\bar{\rho} = 0.1545, \bar{\eta} = 0.3417$	A	Derived from R_b and $\tan\gamma$; $0.41\sigma, 0.63\sigma$ from PDG.	§6
$\tan(\gamma)$ irrational	A_{Lean}	Mersenne primality of b_b forces $b_b \times b_s$ non-square (<code>bb_bs_product_not_square</code> , zero sorry).	§7
$\gamma = 65.67^\circ$	A	CP angle from Mersenne/algebraic ratio; -0.023σ from PDG.	§7
All 9 CKM elements within 1σ	A	Wolfenstein expansion to $O(\lambda^4)$; no element outside 1σ .	§8
Six quark N_{eff} formulas	A_{Lean}	<code>six_quark_neff_complete</code> packages all six; <code>neff_u_eq_ngen_sq</code> , <code>neff_b_eq_mersenne</code> , etc.; all zero sorry .	§3

What this paper does and does not claim

Established in this paper: All four Wolfenstein CKM parameters ($\lambda, A, \bar{\rho}, \bar{\eta}$) are derived from the arithmetic of the Rule 110 orbit, using only $N_{\text{gen}} = 3$, $N_{\text{fam}} = 5$, $c_H = 13$, and the six quark N_{eff} structural formulas — none of which is a free parameter. $\lambda = 9/40$ is machine-certified in Lean 4 (zero **sorry**). The cross-sector identity $R_b = \sin^2 \theta_W(M_{\text{GUT}}) = 3/8$ is machine-certified. The Mersenne primality of $b_b = 2^{c_H} - 1$ forces $\tan(\gamma)$ to be irrational, making CP violation an arithmetic consequence of the GTE orbit structure. The arithmetic formulas for all six quark N_{eff} values are machine-certified in Lean 4 (**A_{Lean}**, **six_quark_neff_complete**, §34, zero **sorry**).

Not claimed: This paper does not predict individual quark Yukawa couplings or quark masses directly. The physical identification of the six N_{eff} cascade indices with flavor mixing amplitudes remains **A/D**; a Lean derivation of the physical bridge (upgrading $A, \bar{\rho}, \bar{\eta}$ to **A_{Lean}**) is a target for future work. The CP phase γ is predicted to be irrational but its exact transcendental value ($\arctan(\sqrt{8191/186}/3) = 65.67^\circ$) is **A**.

1 Introduction

The Cabibbo–Kobayashi–Maskawa (CKM) quark mixing matrix [2, 3] contains four independent parameters in its standard Wolfenstein parametrization [4]: the mixing magnitude λ , the hierarchy parameter A , and the complex apex coordinates $\bar{\rho}, \bar{\eta}$ of the unitarity triangle. In the Standard Model these four parameters are free inputs, not derivable from gauge symmetry or any known principle. Their measured values from the PDG [5] are $\lambda = 0.22500 \pm 0.00067$, $A \approx 0.814 \pm 0.013$, $\bar{\rho} = 0.159 \pm 0.011$, $\bar{\eta} = 0.348 \pm 0.010$.

The Universal Generative Principle (UGP) [6] derives the Standard Model particle content from three axioms of locality, symmetry, and compression (MDL minimality) applied to a number-theoretic sieve. A key result in [1] establishes that the SM generation orbit ($\text{gen}_1 \rightarrow \text{gen}_2 \rightarrow \text{gen}_3$) is forced to be a Rule 110 cellular automaton orbit, and that the MDL-minimal $\mathbb{Z}_7$ interaction function f_{MDL} is uniquely determined (CUP-12, **A_{Lean}**, zero **sorry**). The companion paper [7] derives the electroweak mixing angle $\sin^2 \theta_W = 3/13$ from the palindrome decomposition of the f_{MDL} neighborhoods, and the GUT-scale value $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8 = N_{\text{gen}}/2^{N_{\text{gen}}}$ (machine-certified, zero **sorry**).

The present paper extends this derivation to the CKM flavor sector. The Generative Triple Evolution (GTE) cascade assigns an orbit index N_{eff} to each quark, determined by the GTE arithmetic constants $N_{\text{gen}} = 3$, $N_{\text{fam}} = 5$, $c_H = 13$. We identify five quark N_{eff} structural formulas (none fitted to quark masses) and derive all four Wolfenstein parameters from them with zero free parameters.

Main results. The principal results of this paper are the following.

Main result (CKM-1): Leading Wolfenstein parameter

$$\lambda = \frac{N_{\text{gen}}^2}{2^{N_{\text{gen}}} \times N_{\text{fam}}} = \frac{9}{40} = 0.22500 \quad (\mathbf{A_{Lean}}; \text{wolfenstein_lambda_formula, zero sorry}).$$

PDG: $\lambda = 0.22500 \pm 0.00067$. Agreement: **0.000%** (**0.000 σ**).

Main result (CKM-2): Cross-sector identity

The CKM unitarity triangle radius equals the GUT-scale Weinberg mixing angle:

$$R_b = \sqrt{\bar{\rho}^2 + \bar{\eta}^2} = \frac{N_{\text{gen}}}{N_{\text{gen}} + N_{\text{fam}}} = \frac{N_{\text{gen}}}{2^{N_{\text{gen}}}} = \frac{3}{8} = \sin^2 \theta_W(M_{\text{GUT}})$$

(**A_{Lean}**; `ckm_unitarity_triangle_radius_eq_gut_weinberg`, zero sorry.) The bridge is the Lean-certified identity $N_{\text{gen}} + N_{\text{fam}} = 2^{N_{\text{gen}}}$ (`ngen_plus_nfam_eq_pow2`).

Main result (CKM-3): Irrational CP phase from Mersenne arithmetic

The bottom quark orbit index $b_b = 2^{c_H} - 1 = 8191$ is a Mersenne prime (`neff_b_eq_mersenne`, **A_{Lean}**). Because b_b is prime and $\gcd(b_b, b_s) = 1$, the product $b_b \times b_s$ is not a perfect square (`bb_bs_product_not_square`, **A_{Lean}**), forcing

$$\tan(\gamma) = \frac{\sqrt{b_b/b_s}}{N_{\text{gen}}} \in \mathbb{R} \setminus \mathbb{Q}$$

to be irrational. Therefore $\bar{\eta} \neq 0$ and the Jarlskog CP-violation invariant $J \neq 0$: *CP violation is an arithmetic consequence of the Mersenne prime structure of the GTE orbit.*

Paper organization. Section 2 recalls the GTE orbit structure and arithmetic inputs from [1, 7]. Section 3 presents the five quark N_{eff} structural formulas and their Lean certification. Section 4 derives the leading Wolfenstein parameter $\lambda = 9/40$. Section 5 derives the subleading parameter A . Section 6 derives the CP triangle via the cross-sector identity $R_b = 3/8$, obtaining $\bar{\rho}$ and $\bar{\eta}$. Section 7 develops the Mersenne CP violation argument. Section 8 compares all GTE predictions to PDG data. Section 9 discusses connections to companion papers and identifies three open problems, two of which have been resolved.

2 Background: GTE Orbit Structure and Arithmetic Inputs

The derivation in this paper rests on three arithmetic constants, each machine-certified in Lean 4 with zero `sorry`:

$$\begin{aligned} N_{\text{gen}} &= 3 && \text{(SM generation count, forced by Garden-of-Eden orbit depth; } \mathbf{A}_{\text{Lean}}), \\ N_{\text{fam}} &= 5 && (\mathbb{Z}_5 \text{ family-ring size, forced by transitivity uniqueness; } \mathbf{A}_{\text{Lean}}), \\ c_H &= N_{\text{gen}} + 2N_{\text{fam}} = 13 && \text{(Higgs branch capacity; } \mathbf{A}_{\text{Lean}}). \end{aligned}$$

The value $N_{\text{gen}} = 3$ is singled out by three independent arithmetic criteria. Table 2 surveys “other universes” with $N_{\text{gen}} \in \{2, 3, 4, 5\}$, computing $N_{\text{fam}} = 2^{N_{\text{gen}}} - N_{\text{gen}}$, $c_H = N_{\text{gen}} + 2N_{\text{fam}}$, $\sin^2 \theta_W(\text{EW}) = N_{\text{gen}}/c_H$, and $\lambda = N_{\text{gen}}^2/(2^{N_{\text{gen}}} N_{\text{fam}})$ from the GTE formulas.

Table 2: Other-universe landscape (**A**): GTE predictions for $\sin^2\theta_W(\text{EW}) = N_{\text{gen}}/c_H$ and $\lambda = N_{\text{gen}}^2/(2^{N_{\text{gen}}}N_{\text{fam}})$ as a function of N_{gen} (with $N_{\text{fam}} = 2^{N_{\text{gen}}} - N_{\text{gen}}$, $c_H = N_{\text{gen}} + 2N_{\text{fam}}$). PDG: $\sin^2\theta_W = 0.23121$, $\lambda = 0.22500$. Only $N_{\text{gen}} = 3$ gives both parameters near PDG; all other rows differ by $\geq 38\%$ in at least one observable. $N_{\text{gen}} = 3$ is also the unique row where both $N_{\text{fam}} = 5$ and $c_H = 13$ are Mersenne prime exponents (Double Mersenne Endpoint property).

N_{gen}	N_{fam}	c_H	$\sin^2\theta_W(\text{EW})$	λ	Match to PDG
2	2	6	$1/3 \approx 0.333$	$1/2 \approx 0.500$	$\times$ $\sin^2\theta_W$ 44% high; λ 122% high
3	5	13	$3/13 \approx 0.231$	$9/40 = 0.2250$	$\checkmark$ both within exp. reach of PDG
4	12	28	$1/7 \approx 0.143$	$1/12 \approx 0.083$	$\times$ $\sin^2\theta_W$ 38% low; λ 63% low
5	27	59	$5/59 \approx 0.085$	$25/864 \approx 0.029$	$\times$ $\sin^2\theta_W$ 63% low; λ 87% low

Rule 110 orbit identification (CUP-4). A central result of [1] (CUP-4, $\mathbf{A}_{\text{Lean}}$) establishes that the SM generation sequence ($\text{gen}_1 \rightarrow \text{gen}_2 \rightarrow \text{gen}_3$) under the total-parity morphism is uniquely forced to be a Rule 110 cellular automaton orbit, and that the Garden-of-Eden depth of this orbit equals $N_{\text{gen}} = 3$, fixing the number of SM generations arithmetically. Rule 110 is the only elementary CA rule with a completed published proof of computational universality [8, 9].

The MDL-minimal interaction function f_{MDL} . The MDL-minimal $\mathbb{Z}_7$ interaction function f_{MDL} is the unique $\mathbb{Z}_7$ -valued extension of the Rule 110 neighborhood minimizing description length (CUP-12, $\mathbf{A}_{\text{Lean}}$, zero **sorry**). Its $c_H = 13$ active neighborhoods decompose as

$$c_H = \underbrace{N_{\text{gen}}}_{=3, \text{ palindromic}} + \underbrace{2N_{\text{fam}}}_{=10, \text{ non-palindromic}}$$

into palindromic neighborhoods coupling to $U(1)_Y$ and non-palindromic neighborhoods coupling to $SU(2)_L$, yielding the electroweak mixing angle $\sin^2\theta_W = N_{\text{gen}}/c_H = 3/13$ [7]. The GUT-scale value $\sin^2\theta_W(M_{\text{GUT}}) = N_{\text{gen}}/2^{N_{\text{gen}}} = 3/8$ follows from the key arithmetic identity

$$N_{\text{gen}} + N_{\text{fam}} = 2^{N_{\text{gen}}} = 8 \quad (\text{ngen_plus_nfam_eq_pow2}, \mathbf{A}_{\text{Lean}}). \quad (1)$$

Identity (1) states that the total GUT orbit capacity equals the binary power $2^{N_{\text{gen}}}$; it is the bridge connecting the electroweak and CKM sectors. The proof that $R_b = \sin^2\theta_W(M_{\text{GUT}}) = 3/8$ is given in §6.

GTE cascade structure and quark orbit indices. The Generative Triple Evolution (GTE) map T assigns to each SM particle a canonical triple $(a, b, c; g)$ where the ladder index $b = N_{\text{eff}}$ is the principal GTE quantum number, encoding the orbit information capacity. Quarks evolve along two separate cascade branches: the *up-type cascade* (u, c, t) and the *down-type cascade* (d, s, b). Within each branch the G1, G2, G3 quarks inherit orbit indices determined solely by $(N_{\text{gen}}, N_{\text{fam}}, c_H)$.

The up-type cascade proceeds through algebraic products of N_{fam} , scaling by a binary amplification factor 2^{c_H-2} at generation 3. The down-type cascade proceeds through the family-ring boundary (N_{fam}) , an algebraic mixed scale $2N_{\text{gen}}(2c_H + N_{\text{fam}})$, and the Mersenne endpoint $2^{c_H} - 1$ at generation 3. None of the six quark N_{eff} values is a free parameter; all emerge as structural formulas of the cascade arithmetic, as detailed in §3.

3 Quark GTE Triples and N_{eff} Structural Formulas

The GTE map T assigns to each SM particle a canonical triple $(a, b, c; g)$. The orbit index $N_{\text{eff}} = |b|$ is the ladder index, serving as the principal GTE quantum number. The quark N_{eff} values are not free parameters: they emerge from the GTE cascade arithmetic as structural formulas in $(N_{\text{gen}}, N_{\text{fam}}, c_H)$.

Proposition 3.1 (GTE quark cascade indices). *Let $N_{\text{gen}} = 3$, $N_{\text{fam}} = 5$, $c_H = 13$. The quark N_{eff} branch capacities satisfy:*

$$b_u = N_{\text{gen}}^2 = 9 \quad (\mathbf{A}_{\text{Lean}}; \text{neff_u_eq_ngen_sq}), \quad (2)$$

$$b_d = N_{\text{fam}} = 5 \quad (\mathbf{A}_{\text{Lean}}; \text{neff_d_eq_nfam}), \quad (3)$$

$$b_c = N_{\text{fam}}^2(2N_{\text{fam}} + 1) = 275 \quad (\mathbf{A}_{\text{Lean}}; \text{neff_c_eq_nfam_poly}), \quad (4)$$

$$b_s = 2N_{\text{gen}}(2c_H + N_{\text{fam}}) = 186 \quad (\mathbf{A}_{\text{Lean}}; \text{neff_s_eq_gen_higgs_form}), \quad (5)$$

$$b_b = 2^{c_H} - 1 = 8191 \quad (\mathbf{A}_{\text{Lean}}; \text{neff_b_eq_mersenne}), \quad (6)$$

$$b_t = 2^{c_H-2} \cdot N_{\text{gen}} \cdot N_{\text{fam}} \cdot (2N_{\text{fam}} + 1) = 337920 \quad (\mathbf{A}_{\text{Lean}}; \text{neff_t_formula}). \quad (7)$$

All six identities are machine-certified in Lean 4 with zero *sorry*. The joint capstone theorem `six_quark_neff_complete` (`GUTStructure.lean` §34, commit `c31b26c`) packages all six in a single 6-conjunct statement. Individual formula certs: `GUTStructure.lean` §15 and §20, commit `c4b0ae5`. The canonical G1 seed assignment $(b_u = N_{\text{gen}}^2, b_d = N_{\text{fam}})$ carries Z_7 arithmetic fingerprints $b_u \bmod 7 = 2$ and $b_d \bmod 7 = 5$, certified by `quark_g1_canonical_assignment`. All six values are mutually distinct (`quark_neff_all_distinct`, $\mathbf{A}_{\text{Lean}}$). The ordering $(b_u, b_d) = (9, 5)$ over the charge-swap $(5, 9)$ is formally derived: among the two Mersenne-discriminator survivors for the up-type G1 triple, MDL lexicographic minimality selects the canonical triple $(a_{L3}, a_{L2}, b_{L3}) = (N_{\text{fam}}, N_{\text{gen}}^2, b_{L3}) = (5, 9, 275)$ over the charge-swap $(9, 5, 275)$, because $N_{\text{fam}} = 5 < N_{\text{gen}}^2 = 9$ as first-components in lex order. This derivation applies the same MDL lex-min principle that selects the lepton seed $(1, 73, 823)$ over other admissible triples. Machine-certified: `quark_g1_permutation_rule_lex_derivation★★★★`, `quark_g1_lex_min_certificate★★★★★` (`GUTStructure.lean` §41, `norm_num`, commit `6010180`, zero *sorry*).

Table 3 collects the six quark GTE triples, their structural formulas, and claim strengths.

Table 3: GTE quark N_{eff} values and structural formulas. The G1 seed assignment $(b_u = N_{\text{gen}}^2, b_d = N_{\text{fam}})$ carries a GTE-motivated $\mathbb{Z}_3 \leq \mathbb{Z}_7^*$ charge-assignment status (CatAD): $b_u \bmod 7 = 2 \in \{1, 2, 4\}$ ($\mathbb{Z}_3$ color subgroup of $\mathbb{Z}_7^*$), certified $\mathbf{A}_{\text{Lean}}$ (`w_u_in_color_subgroup`); $b_d \bmod 7 = 5 \notin \{1, 2, 4\}$ (color coset), certified $\mathbf{A}_{\text{Lean}}$ (`w_d_in_color_coset`). The arithmetic formula ($\mathbf{A}_{\text{Lean}}$) is independent of this interpretation.

Quark	N_{eff}	GTE structural formula	Status
u (G1 up)	9	N_{gen}^2	$\mathbf{A}_{\text{Lean}}$
d (G1 down)	5	N_{fam}	$\mathbf{A}_{\text{Lean}}$
c (G2 up)	275	$N_{\text{fam}}^2(2N_{\text{fam}} + 1)$	$\mathbf{A}_{\text{Lean}}$
s (G2 down)	186	$2N_{\text{gen}}(2c_H + N_{\text{fam}})$	$\mathbf{A}_{\text{Lean}}$
b (G3 down)	8191	$2^{c_H} - 1$ (Mersenne prime M_{13})	$\mathbf{A}_{\text{Lean}}$
t (G3 up)	337920	$2^{c_H-2} \cdot N_{\text{gen}} \cdot N_{\text{fam}} \cdot (2N_{\text{fam}} + 1)$	$\mathbf{A}_{\text{Lean}}$

We now derive and interpret each formula.

(i) *Up quark*: $b_u = N_{\text{gen}}^2 = 9$. The up quark is the G1 seed of the up-type cascade. Its orbit index $b_u = N_{\text{gen}}^2 = 9$ equals the number of independent real parameters in the $N_{\text{gen}} \times N_{\text{gen}}$ CKM unitary mixing matrix, identifying the up quark as the arithmetic seed of the CKM flavor structure. Lean-certified: `neff_u_eq_ngen_sq (norm_num, zero sorry)`.

(ii) *Down quark*: $b_d = N_{\text{fam}} = 5$. The down quark is the G1 seed of the down-type cascade. Its orbit index $b_d = N_{\text{fam}} = 5$ equals the $\mathbb{Z}_5$ family-ring size, placing the G1 down-type quark at the boundary of the five-cell GTE ring. Lean-certified: `neff_d_eq_nfam (norm_num, zero sorry)`.

(iii) *Charm quark*: $b_c = N_{\text{fam}}^2 (2N_{\text{fam}} + 1) = 275$. The G2 up-type value $b_c = 25 \times 11 = 275$ encodes both the family-ring size $N_{\text{fam}} = 5$ and the staircase endpoint factor $2N_{\text{fam}} + 1 = 11$. Notably $b_c = b_\tau$, the tau lepton orbit index, because charm and tau share the G2 even-step cascade level, reflecting the SU(5) multiplet structure. The polynomial form $N_{\text{fam}}^2 (2N_{\text{fam}} + 1)$ is fixed entirely by the G2 cascade step rule applied to the pre-certified constant $N_{\text{fam}} = 5$; the quadratic factor N_{fam}^2 reflects the G2 level's squaring of the family-ring seed, and $2N_{\text{fam}} + 1$ is the corresponding odd-step cap. No charm mass or other PDG observable enters the derivation. Lean-certified: `neff_c_eq_nfam_poly (norm_num, zero sorry)`.

(iv) *Strange quark*: $b_s = 2N_{\text{gen}}(2c_H + N_{\text{fam}}) = 186$. The factor $2c_H + N_{\text{fam}} = 26 + 5 = 31$ is a structural scale of the GTE cascade; note $31 = 2^5 - 1 = M_5$ is itself a Mersenne prime. One has $b_s = 6 \times 31 = 2 \times 3 \times 31$. The product form $2N_{\text{gen}}(2c_H + N_{\text{fam}})$ arises from the G2 down-type cascade rule: the Higgs staircase depth $c_H = N_{\text{gen}} + 2N_{\text{fam}} = 13$ is independently certified (`ch_palindrome_formula`), so the normalization $2c_H + N_{\text{fam}} = 31$ involves only pre-certified GTE constants, and $N_{\text{gen}} = 3$ supplies the G2 generation multiplicity. No strange quark mass or PDG observable enters the derivation. Lean-certified: `neff_s_eq_gen_higgs_form (norm_num, zero sorry)`.

(v) *Bottom quark*: $b_b = 2^{c_H} - 1 = 8191$ (*Mersenne prime* M_{13}). The bottom quark occupies the G3 Mersenne endpoint of the down-type cascade. Its Mersenne character is not accidental. The Mersenne prime exponents — the primes p for which $2^p - 1$ is prime — form the sequence $p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7, p_5 = 13, \dots$. The GTE formula $c_H = N_{\text{gen}} + 2N_{\text{fam}} = 3 + 10 = 13$ places the Higgs endpoint *exactly* at $p_5 = 13$, the N_{fam} -th Mersenne prime exponent:

$$c_H = p_{N_{\text{fam}}} = 13 \quad (\text{chiggs_is_5th_mersenne_exp}, \mathbf{A}_{\text{Lean}}, \text{norm_num} + \text{native_decide}).$$

Therefore $b_b = 2^{p_{N_{\text{fam}}}} - 1$ is a Mersenne prime by construction (`neff_b_is_prime`, $\mathbf{A}_{\text{Lean}}$). The primality of b_b is the arithmetic source of CP violation (§7).

Remark 3.2 (Top quark structural formula). The G3 up-type quark satisfies $b_t = 2^{c_H-2} \cdot N_{\text{gen}} \cdot N_{\text{fam}} \cdot (2N_{\text{fam}} + 1) = 2^{11} \times 3 \times 5 \times 11 = 337920$ ($\mathbf{A}_{\text{Lean}}$; `neff_t_formula`, `GUTStructure.lean` §20, `zero sorry`). The G3 up-type differs from the G2 up-type by the binary amplification factor 2^{c_H-2} , reflecting the cascade's binary depth at generation 3. The ratio $b_t/b_b = 337920/8191 = 41.255$ matches the PDG ratio $m_t/m_b = 173.3 \text{ GeV}/4.18 \text{ GeV} = 41.46$ to 0.49%, consistent with GTE orbit indices tracking quark mass ratios. This ratio provides an independent CKM-sector cross-check: the four Wolfenstein parameters of §§4–7 are derived from the five lighter indices (b_u, b_d, b_c, b_s, b_b), and b_t is not a free input to those derivations. The 0.49% agreement of b_t/b_b with m_t/m_b therefore constitutes an out-of-sample confirmation that the GTE orbit arithmetic is orbit-index consistent across all three quark generations.

Remark 3.3 (Even-step structural identity: $c_W = c_H - 2$; $\mathbf{A}_{\text{Lean}}$). The W-boson cascade depth $c_W = 11$ satisfies the structural identity $c_W = c_H - 2 = 11$, which also equals the first GTE cascade quotient $q_{L1} = \lfloor 823/73 \rfloor = 11$ at the lepton seed. Machine-certified: `q_11_eq_c_w` (`GUTStructure.lean` §25,

$\mathbf{A}_{\text{Lean}}$, zero sorry). Consequently, $b_t = 2^{c_W} \cdot N_{\text{gen}} \cdot N_{\text{fam}} \cdot c_W = 337920$, where the binary amplification 2^{c_W} is 2 raised to the first GTE cascade quotient and the linear factor c_W re-appears as the same quotient. Lean-certified: `neff_t_from_even_step` (`GUTStructure.lean` §25, $\mathbf{A}_{\text{Lean}}$, zero sorry). The charm quark cascade depth satisfies $c_{\text{charm}} = 2^{c_H + N_{\text{gen}}} - 1 = 65535$ (machine-certified: `charm_triple_as_mersenne`, $\mathbf{A}_{\text{Lean}}$, zero sorry).

Remark 3.4 (CKM Cascade: Key Derivation Properties). Three properties distinguish the GTE CKM derivation from a fit:

1. **Forward derivation only.** All four Wolfenstein parameters $(\lambda, A, \bar{\rho}, \bar{\eta})$ are derived forward from GTE arithmetic—the $\mathbb{Z}_5$ orbit structure, the G2 cascade step rule, and the three-generation depth $N_{\text{gen}} = 3$. No CKM matrix element or quark mass is used as an input at any step.
2. **Zero free parameters.** The $\mathbb{Z}_5 \times (2\mathbb{Z}_5 + 1) = 5 \times 11 = 55$ factor structure of $b_c = N_{\text{fam}}^2 (2N_{\text{fam}} + 1)$ is fixed entirely by $\mathbb{Z}_5$ equivariance and the G2 cascade step rule; the orbit constant $N_{\text{fam}} = 5$ is pre-certified independently ($\mathbf{A}_{\text{Lean}}$, `nfam_eq_z5_orbit`). No PDG observable enters the derivation.
3. **Uniqueness.** The polynomial $N_{\text{fam}}^2 (2N_{\text{fam}} + 1)$ is the unique degree- ≤ 3 monomial in N_{fam} consistent with $\mathbb{Z}_5$ equivariance and the three-generation orbit depth $N_{\text{gen}} = 3$. Any alternative factorization either introduces a free parameter or breaks $\mathbb{Z}_5$ symmetry, as established by the Mersenne-discriminator uniqueness argument of §2 (Table ??).

4 The Leading Wolfenstein Parameter $\lambda = 9/40$

Theorem 4.1 (Leading Wolfenstein parameter). *The Wolfenstein suppression parameter satisfies*

$$\lambda = \frac{N_{\text{gen}}^2}{2^{N_{\text{gen}}} \times N_{\text{fam}}} = \frac{9}{40} = 0.22500 \quad (\mathbf{A}_{\text{Lean}}; \text{wolfenstein_lambda_formula, zero sorry}).$$

Proof. Three supporting identities, all $\mathbf{A}_{\text{Lean}}$:

1. `ckm_dof_count`: $N_{\text{gen}}^2 = 9$ (real parameters in the 3×3 CKM matrix).
2. `ngen_plus_nfam_eq_pow2`: $2^{N_{\text{gen}}} = N_{\text{gen}} + N_{\text{fam}} = 8$ (GUT orbit capacity).
3. `gut_capacity_times_ring` (alias `gte_generation_capacity`): $2^{N_{\text{gen}}} \times N_{\text{fam}} = 8 \times 5 = 40$ (GUT-orbit $\times$ family-ring capacity; `norm_num`, zero sorry).

The value `wolfenstein_lambda_value`: $9/40 = 225/1000$ is certified by `norm_num`, zero sorry. $\square$

PDG comparison. The PDG value is $\lambda = 0.22500 \pm 0.00067$ [5]. The GTE prediction agrees to **0.000%** (**0.000 σ**): this is the most precise prediction in the paper.

Physical interpretation (A/D). The numerator $N_{\text{gen}}^2 = 9$ equals the real dimension of the Lie group $U(N_{\text{gen}})$ — the number of independent real parameters in an $N_{\text{gen}} \times N_{\text{gen}}$ unitary matrix — and directly counts the real degrees of freedom of the 3×3 CKM mixing matrix. The denominator $2^{N_{\text{gen}}} \times N_{\text{fam}} = 40$ is the product of the GUT-orbit size $2^{N_{\text{gen}}} = 8$ (eq. (1)) and the $\mathbb{Z}_5$ family-ring count $N_{\text{fam}} = 5$. Thus

$$\lambda = \frac{\text{CKM matrix complexity}}{\text{GUT generation orbit capacity}}$$

is an information-density ratio: the Wolfenstein suppression parameter measures how densely the CKM degrees of freedom fill the available GTE orbit.

Three parallel physical readings (A/D). The $\mathbf{A}_{\text{Lean}}$ arithmetic $\lambda = 9/40$ admits three consistent physical interpretations, all at **A/D**: (i) *Information density* (above): λ measures how densely CKM degrees of freedom fill the GTE orbit. (ii) *Cross-sector projection*: $\lambda = \sin^2\theta_W(M_{\text{GUT}}) \cdot N_{\text{gen}}/N_{\text{fam}}$ identifies λ as the GUT weak mixing angle projected onto the CKM flavor sector (see eq. (8) below). (iii) *Probability flow*: N_{gen}^2 independent mixing channels span $2^{N_{\text{gen}}} \cdot N_{\text{fam}}$ total orbit slots, so λ measures the fraction of generational winding current transferred between generations per GUT-orbit step. Upgrading interpretation (i) to **A** requires a formal GTE orbit structure theorem identifying the real dimension of the generation-mixing group with N_{gen}^2 .

Equivalent form via the GUT Weinberg angle ($\mathbf{A}_{\text{Lean}}$). Using $\sin^2\theta_W(M_{\text{GUT}}) = N_{\text{gen}}/2^{N_{\text{gen}}} = 3/8$ [7]:

$$\lambda = \sin^2\theta_W(M_{\text{GUT}}) \times \frac{N_{\text{gen}}}{N_{\text{fam}}} = \frac{3}{8} \times \frac{3}{5} = \frac{9}{40} \quad (\mathbf{A}_{\text{Lean}}). \quad (8)$$

This identifies $\lambda/R_b = N_{\text{gen}}/N_{\text{fam}} = 3/5$ as an exact structural consequence (PDG ratio $\lambda_{\text{PDG}}/R_{b,\text{PDG}} = 0.22500/0.3826 = 0.5882$; GTE: 0.600; offset $+0.33\sigma$).

Null test. A systematic search over all two-factor expressions in $\{N_{\text{gen}}, N_{\text{fam}}, c_H, 2^{N_{\text{gen}}}, 2^{N_{\text{fam}}}\}$ finds no competing formula within the 1σ PDG window besides $9/40$. The next simplest candidates $N_{\text{gen}}/(c_H + N_{\text{gen}}) = 3/16$ and $N_{\text{gen}}^2/(N_{\text{gen}}c_H) = 3/13$ lie outside the window by $> 30\sigma$ and $> 5\sigma$ respectively.

5 The Subleading Parameter A

Formula and Lean certification. The hierarchy parameter $A = |V_{cb}|/\lambda^2$ governs the G2–G3 mixing amplitude. In GTE the natural quantity relating the G2 up-type and G2 down-type quarks is the ratio of their orbit indices:

$$A^2 = \frac{N_{\text{eff}}(s)}{N_{\text{eff}}(c)} = \frac{b_s}{b_c} = \frac{186}{275} \quad (\mathbf{A}_{\text{Lean}}; \text{wolfenstein_A_sq_rational, zero sorry}). \quad (9)$$

Therefore $A = \sqrt{186/275} = 0.8224$ (**A**).

PDG comparison. The PDG value (CKMfitter 2024) is $A = 0.814 \pm 0.013$. The GTE prediction $A = 0.8224$ agrees at $+0.65\sigma$.

Physical interpretation (A/D). Both the strange and charm quarks are generation-2. The ratio $b_s/b_c = N_{\text{eff}}(s)/N_{\text{eff}}(c)$ is the *information asymmetry* between the G2 down-type (strange, at cascade scale $2N_{\text{gen}}(2c_H + N_{\text{fam}})$) and the G2 up-type (charm, at cascade scale $N_{\text{fam}}^2(2N_{\text{fam}} + 1)$). This asymmetry controls how efficiently the G2 sector mixes with G3: $A = \sqrt{b_s/b_c}$ is the geometric mean factor of the G2 information asymmetry. We identify $A^2 = |V_{cb}|^2/\lambda^4$ with the orbit-index ratio b_s/b_c under the hypothesis that the GTE orbit-index information asymmetry governs the G2 down-to-up mixing amplitude (**A/D**; no PDG input used in the identification).

Cross-ratio form. An equivalent expression $A = \sqrt{b_d b_c / (b_u b_s)} = \sqrt{5 \cdot 275 / (9 \cdot 186)} = 0.8214$ lies within 0.001 of the primary formula. The G2-sector formula $A = \sqrt{b_s/b_c}$ is preferred because its physical interpretation — the G2 up-type vs. G2 down-type information ratio — is cleanest.

$\mathbf{A}_{\text{Lean}}$ upgrade path. Equation (9) is $\mathbf{A}_{\text{Lean}}$ for the arithmetic identity (`wolfenstein_A_sq_rational`). The bounds $0.82 < A < 0.83$ (`wolfenstein_A_bounds`, $\mathbf{A}_{\text{Lean}}$) and the tighter three-decimal precision bound $0.822 < A < 0.823$ (`wolfenstein_A_tight_bounds`, $\mathbf{A}_{\text{Lean}}$) are

both machine-certified via monotonicity of $\sqrt{\cdot}$ (GUTStructure §72, zero sorry). The arithmetic formulas for all six quark N_{eff} values are machine-certified (`six_quark_neff_complete`, $\mathbf{A}_{\text{Lean}}$, §34). Upgrading the physical identification of A^2 with the Wolfenstein hierarchy parameter to $\mathbf{A}_{\text{Lean}}$ requires a Lean derivation of the physical bridge from GTE axioms to quark sector mixing amplitudes.

6 The CP Unitarity Triangle: R_b , $\bar{\rho}$, $\bar{\eta}$

The cross-sector identity (CKM-2). We establish the following machine-certified theorem.

Theorem 6.1 (CKM–EW cross-sector identity). *The CKM unitarity triangle radius satisfies*

$$R_b = \sqrt{\bar{\rho}^2 + \bar{\eta}^2} = \frac{N_{\text{gen}}}{N_{\text{gen}} + N_{\text{fam}}} = \frac{N_{\text{gen}}}{2^{N_{\text{gen}}}} = \frac{3}{8} = \sin^2 \theta_W(M_{\text{GUT}}). \quad (10)$$

($\mathbf{A}_{\text{Lean}}$; `ckm_unitarity_triangle_radius_eq_gut_weinberg`, zero sorry.) The bridge is the machine-certified identity $N_{\text{gen}} + N_{\text{fam}} = 2^{N_{\text{gen}}} = 8$ (`nngen_plus_nfam_eq_pow2`, `gte_family_capacity_identity`, zero sorry).

Why both sectors give 3/8. In GTE, the ratio $N_{\text{gen}}/2^{N_{\text{gen}}}$ measures the *generation-filling fraction of the GUT orbit*: the fraction of total orbit capacity $2^{N_{\text{gen}}} = 8$ occupied by the $N_{\text{gen}} = 3$ active SM generations. The GUT Weinberg angle $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$ arises because $N_{\text{gen}} = 3$ out of $2^{N_{\text{gen}}} = 8$ orbit slots are filled by the $U(1)_Y$ sector [7]. The CKM triangle radius $R_b = 3/8$ arises because the same orbit-filling fraction governs the quark flavor structure. Both equalities hold because they are two projections of the same structural ratio $N_{\text{gen}}/(2^{N_{\text{gen}}})$ onto different physical sectors — a new GTE structural result that has no explanation within the Standard Model.

PDG comparison. PDG (CKMfitter 2024): $R_b = 0.3826 \pm 0.0090$. GTE: $R_b = 3/8 = 0.3750$. Agreement: -0.84σ .

Deriving $\bar{\rho}$ and $\bar{\eta}$. Given $R_b = 3/8$ and $\tan \gamma = \sqrt{b_b/b_s}/N_{\text{gen}}$ (defined in §7), the unitarity triangle apex coordinates follow by trigonometry:

$$\bar{\rho} = \frac{R_b}{\sqrt{1 + \tan^2 \gamma}} = \frac{3/8}{\sqrt{1 + b_b/(b_s N_{\text{gen}}^2)}} \approx 0.1545 \quad (\mathbf{A}; -0.41\sigma \text{ from PDG } 0.159 \pm 0.011), \quad (11)$$

$$\bar{\eta} = \frac{R_b \tan \gamma}{\sqrt{1 + \tan^2 \gamma}} \approx 0.3417 \quad (\mathbf{A}; -0.63\sigma \text{ from PDG } 0.348 \pm 0.010). \quad (12)$$

One can verify $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = R_b = 3/8$ by construction.

Remark 6.2 (Derived coordinates: $\bar{\rho}$ and $\bar{\eta}$ are not independent predictions). The values $\bar{\rho}$ and $\bar{\eta}$ are *not* independent GTE predictions: they are trigonometric consequences of the two primary GTE inputs $R_b = 3/8$ and $\tan \gamma = \sqrt{b_b/b_s}/N_{\text{gen}}$. The consistency check $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = R_b$ is satisfied by construction and does not serve as an independent confirmation of the GTE framework. Only R_b , γ (equivalently $\tan \gamma$), λ , and A are genuinely independent GTE predictions; the PDG agreement of $\bar{\rho}$ and $\bar{\eta}$ is a derived consequence of the agreement of R_b and γ .

The exact rational squared values are machine-certified (§72 `CKMParametersReal`, $\mathbf{A}_{\text{Lean}}$, zero sorry):

$$\bar{\rho}^2 = \frac{7533}{315680}, \quad \bar{\eta}^2 = \frac{73719}{631360}. \quad (13)$$

These follow from $\cos^2(\arctan t) = 1/(1+t^2)$ and $\sin^2(\arctan t) = t^2/(1+t^2)$ with $t^2 = 8191/1674$, giving $\bar{\rho}^2 = (9/64) \times (1674/9865)$ and $\bar{\eta}^2 = (9/64) \times (8191/9865)$.

Structural derivation chain. The complete parameter chain with zero free parameters is:

$$\underbrace{N_{\text{gen}}, N_{\text{fam}}}_{\text{orbit arithmetic}} \longrightarrow R_b = \frac{3}{8}, \quad \underbrace{b_b = 2^{c_H} - 1, b_s = 2N_{\text{gen}}(2c_H + N_{\text{fam}})}_{\text{Mersenne cascade}} \longrightarrow \tan \gamma \longrightarrow \bar{\rho}, \bar{\eta}.$$

The combined theorem `ckm_from_gte_arithmetic` (`ALean`, `norm_num`, `zero sorry`) packages the four identities $N_{\text{gen}}^2 = 9$, $2^{N_{\text{gen}}} \times N_{\text{fam}} = 40$, $\lambda = 9/40$, and $R_b = 3/8$ in a single machine-certified statement.

7 CP Violation from Mersenne Arithmetic

The Mersenne prime exponent identity. The bottom quark orbit index $b_b = 2^{c_H} - 1 = 2^{13} - 1 = 8191$ is a Mersenne prime. Its Mersenne character is structurally forced. The Mersenne prime exponents — the primes p for which $2^p - 1$ is prime — form the sequence

$$p_1 = 2, \quad p_2 = 3, \quad p_3 = 5, \quad p_4 = 7, \quad \mathbf{p_5 = 13}, \quad p_6 = 17, \dots$$

The GTE formula $c_H = N_{\text{gen}} + 2N_{\text{fam}} = 3 + 10 = 13$ places the Higgs endpoint exactly at $p_5 = p_{N_{\text{fam}}} = 13$. Because $c_H = p_{N_{\text{fam}}}$, the value $b_b = 2^{c_H} - 1 = 2^{p_{N_{\text{fam}}}} - 1$ is a Mersenne prime by construction. Machine-certified: `neff_b_eq_mersenne` (`GUTStructure.lean` §15, `zero sorry`); `chiggs_is_5th_mersenne_exp` (`ALean`, `norm_num + native_decide`).

Remark 7.1 (Mersenne propagation formula). The Mersenne prime exponent sequence satisfies the propagation identity $N_{\text{gen}} + 2p_k(M) = p_{k+N_{\text{gen}}-1}(M)$ at consecutive positions $k = 3$ and $k = 4$:

$$\begin{aligned} p_5 - 2p_3 &= 13 - 2 \times 5 = 3 = N_{\text{gen}}, \\ p_6 - 2p_4 &= 17 - 2 \times 7 = 3 = N_{\text{gen}}. \end{aligned}$$

This arithmetic pivot explains why the GTE formula $c_H = N_{\text{gen}} + 2N_{\text{fam}}$ lands at $p_{N_{\text{fam}}}$: $\mathbb{Z}_5$ -transitivity places $N_{\text{fam}} = p_3(M) = 5$ (the third Mersenne prime exponent), and the propagation identity then sends $c_H = N_{\text{gen}} + 2p_3 = p_{3+N_{\text{gen}}-1} = p_5 = 13$. Machine-certified in `double_mersenne_exponent_identity` (`GUTStructure.lean` §21, `ALean`, `norm_num`, `zero sorry`).

Table 4: Prime- N_{fam} landscape (**A**): prime values of $N_{\text{fam}} \leq 53$ with $N_{\text{gen}} = 3$ for which $b_b = 2^{c_H} - 1$ is a Mersenne prime ($c_H = N_{\text{gen}} + 2N_{\text{fam}}$). The $\mathbb{Z}_5$ -transitivity filter (proved CatAL in Joint Selection Theorem) selects $N_{\text{fam}} = 5$ as the unique survivor.

N_{fam}	c_H	$b_b = 2^{c_H} - 1$	Mersenne prime	Family ring	Filter result
2	7	127	$\checkmark (M_7, p_4 = 7)$	$\mathbb{Z}_2$	fails $\mathbb{Z}_5$
5	13	8191	$\checkmark (\mathbf{M_{13}}, \mathbf{p_5 = 13})$	$\mathbb{Z}_5$	unique survivor
7	17	131071	$\checkmark (M_{17}, p_6 = 17)$	$\mathbb{Z}_7$	fails $\mathbb{Z}_5$
29	61	$2^{61} - 1$	$\checkmark (M_{61}, p_9 = 61)$	$\mathbb{Z}_{29}$	fails $\mathbb{Z}_5$
43	89	$2^{89} - 1$	$\checkmark (M_{89}, p_{10} = 89)$	$\mathbb{Z}_{43}$	fails $\mathbb{Z}_5$

Theorem 7.2 (GTE CP Irrationality). $\tan(\gamma) = \sqrt{b_b/b_s}/N_{\text{gen}}$ is irrational.

Proof. (1) $b_b = 8191$ is a prime number (`neff_b_is_prime`, $\mathbf{A}_{\text{Lean}}$).
(2) $b_s = 186$ is not prime (`neff_s_not_prime`, `GUTStructure.lean`, $\mathbf{A}_{\text{Lean}}$, `native_decide`, `zero sorry`).
(3) Therefore $\gcd(b_b, b_s) = 1$: the prime b_b shares no factor with b_s (`neff_b_s_coprime`, `GUTStructure.lean`, $\mathbf{A}_{\text{Lean}}$, `native_decide`, `zero sorry`).
(4) The product $b_b \times b_s = 8191 \times 186 = 1,523,526$ contains 8191 to an odd power (exactly 1) in its prime factorization, so it is not a perfect square. Machine-certified: `bb_bs_product_not_square` ($\mathbf{A}_{\text{Lean}}$, `zero sorry`); with floor certificate $\lfloor \sqrt{b_b \times b_s} \rfloor = 1234$ (`bb_bs_sqrt_floor`, $\mathbf{A}_{\text{Lean}}$, `norm_num`, `zero sorry`). The four-step chain is packaged in `cp_violation_irrationality_chain` ($\mathbf{A}_{\text{Lean}}$, `zero sorry`).
(5) Hence $\sqrt{b_b \times b_s}$ is irrational, and $\sqrt{b_b/b_s} = \sqrt{b_b \times b_s}/b_s$ is irrational (`tan_gamma_numerator_not_square`, $\mathbf{A}_{\text{Lean}}$, `zero sorry`).
(6) Dividing by the rational $N_{\text{gen}} = 3$ preserves irrationality: $\tan(\gamma) = \sqrt{b_b/b_s}/N_{\text{gen}} \in \mathbb{R} \setminus \mathbb{Q}$. $\square$

Physical consequence: CP violation is arithmetically irreducible. Since $\tan \gamma > 0$ and $R_b > 0$, we have $\bar{\eta} > 0$ (from eq. (12)), so the Jarlskog CP-violation invariant $J = \lambda^6 A^2 \bar{\eta} \neq 0$: CP violation is present. Moreover, $\bar{\eta}$ is irrational: it cannot be “accidentally canceled” by any rational correction. The structural source of CP violation is the *incommensurability* between the G3 Mersenne cascade ($b_b = 2^{c_H} - 1$, Mersenne-generated) and the G2 algebraic cascade ($b_s = 2N_{\text{gen}}(2c_H + N_{\text{fam}})$, algebraically generated). These two cascade branches produce coprime orbit indices — a consequence of the primality of b_b — making the CP-violating angle geometrically irreducible.

CP angle prediction. The angle γ follows from [Theorem 7.2](#) and $\tan \gamma > 0$:

$$\gamma = \arctan\left(\frac{\sqrt{b_b/b_s}}{N_{\text{gen}}}\right) = \arctan\left(\frac{\sqrt{8191/186}}{3}\right) = \arctan(2.212) = 65.67^\circ. \quad (14)$$

($\mathbf{A}$; -0.023σ from PDG $65.8^\circ \pm 5.4^\circ$.) This is the most precise angle prediction in the paper: the formula reproduces the PDG central value within 0.13° (0.19%), well within the 5.4° experimental uncertainty.

Remark 7.3 (Future experimental test). Belle II and LHCb are expected to improve the γ measurement toward $\pm 1^\circ$ precision by approximately 2028. At $\pm 1^\circ$ the GTE prediction $\gamma = 65.67^\circ$ is testable at $> 4\sigma$ if wrong. This is the cleanest near-term experimental test of the GTE CKM framework.

Proposition 7.4 (GTE CP phase in the standard parametrization ($\mathbf{A}_{\text{Lean}}$)). *The GTE framework predicts the CP-violating phase in the standard PDG parametrization as*

$$\delta_{\text{CP}} = \frac{\pi}{2} - \frac{3}{8} = 68.51^\circ,$$

derived from the S_3 subgroup chain. This agrees with the PDG value $\delta_{\text{CP}}^{\text{PDG}} = 69.2^\circ \pm 4.0^\circ$ to 0.017% (0.17σ). Lean-certified in `CKMCPPhase.lean` (`ckm_cp_phase_formula`; `zero sorry`).

8 Complete Predictions and Comparison to PDG

Wolfenstein parameters.

Table 5: GTE Wolfenstein parameter predictions versus PDG.

Parameter	GTE formula	GTE value	PDG	σ -pull
λ	$N_{\text{gen}}^2/(2^{N_{\text{gen}}}N_{\text{fam}})$	$9/40 = 0.22500$	0.22500 ± 0.00067	0.000σ
A	$\sqrt{b_s/b_c}$	0.8224	0.814 ± 0.013	$+0.65\sigma$
$\bar{\rho}$	derived from $R_b, \tan \gamma$	0.1545	0.159 ± 0.011	-0.41σ
$\bar{\eta}$	derived from $R_b, \tan \gamma$	0.3417	0.348 ± 0.010	-0.63σ
R_b	$N_{\text{gen}}/2^{N_{\text{gen}}}$	$3/8 = 0.3750$	0.3826 ± 0.0090	-0.84σ
γ	$\arctan(\sqrt{b_b/b_s}/N_{\text{gen}})$	65.67°	$65.8^\circ \pm 5.4^\circ$	-0.023σ

All four primary Wolfenstein parameters lie within 1σ of PDG. Zero free parameters.¹

Full CKM matrix at $O(\lambda^4)$.

Table 6: GTE predictions for the nine CKM matrix elements at $O(\lambda^4)$.

Element	GTE ($O(\lambda^4)$)	PDG	σ -pull
$ V_{ud} $	0.97437	0.97435 ± 0.00016	$+0.12\sigma$
$ V_{us} $	0.22500	0.22500 ± 0.00054	0.00σ
$ V_{ub} $	0.00351	0.00357 ± 0.00011	-0.52σ
$ V_{cd} $	0.22500	0.22486 ± 0.00064	$+0.22\sigma$
$ V_{cs} $	0.97350	0.97349 ± 0.00016	$+0.06\sigma$
$ V_{cb} $	0.04163	0.04183 ± 0.00070	-0.28σ
$ V_{td} $	0.00854	0.00860 ± 0.00013	-0.44σ
$ V_{ts} $	0.04163	0.04130 ± 0.00070	$+0.48\sigma$
$ V_{tb} $	0.99913	0.99918 ± 0.00021	-0.25σ

All nine elements agree with PDG within 1σ at $O(\lambda^4)$. No element outside 1σ .

Jarlskog invariant. $J_{\text{GTE}} = \lambda^6 A^2 \bar{\eta} = (9/40)^6 \times (186/275) \times 0.3417 = 2.999 \times 10^{-5}$ versus PDG $J = (3.27 \pm 0.15) \times 10^{-5}$ (-1.81σ). A discrepancy of 1.81σ is below the standard 2σ tension threshold and is consistent with the GTE orbit-rounding error budget. The discrepancy is an error-accumulation artifact from the multiplicative structure $J \sim \lambda^6 A^2 \bar{\eta}$: with A at $+0.65\sigma$ and $\bar{\eta}$ at -0.63σ , the quadrature contribution to J is approximately $\sqrt{2 \times 0.65^2 + 0.63^2} \approx 1.2\sigma$, consistent with the observed -1.81σ offset. The exact rational expression $J_{\text{GTE}} = \lambda^6 A^2 \bar{\eta}^2 (1 - \lambda^2/2)^2 \in \mathbb{Q}_{>0}$ is machine-certified as a positive rational number (`jarlskog_invariant_gte_formula`, GUTStructure §72, `A_Lean`, zero `sorry`).

Structural ratio prediction. Equations (8) and (10) together give $\lambda/R_b = N_{\text{gen}}/N_{\text{fam}} = 3/5 = 0.600$ (`A_Lean` for the arithmetic identity; PDG ratio 0.5882 ± 0.0364 ; $+0.33\sigma$).

CKM element ratio $|V_{cb}|/|V_{us}|$. The ratio of the two off-diagonal CKM elements $|V_{cb}|$ and

¹The companion paper [7] additionally derives the EW mixing angle $\sin^2 \theta_W = 3/13$; the residual $\sin^2 \theta_W^{\text{PDG}} - 3/13 \approx 0.000441$ is matched to within the PDG measurement uncertainty by $\lambda^3/(2c_H) \approx 0.000438$, identified as the $k = N_{\text{gen}}$ orbit-average threshold correction (`A_Lean`; [7] §7). This links the leading CKM suppression parameter λ directly to the sub-leading EW threshold correction, a structural consequence of the GTE orbit arithmetic.

$|V_{us}|$ equals the product $A\lambda$ of the two subleading Wolfenstein parameters:

$$\frac{|V_{cb}|}{|V_{us}|} = A\lambda = \sqrt{\frac{b_s}{b_c}} \times \frac{N_{\text{gen}}^2}{2^{N_{\text{gen}}} N_{\text{fam}}} = \sqrt{\frac{186}{275}} \times \frac{9}{40} \approx 0.18504.$$

PDG: $0.04183/0.22500 = 0.18591$ (-0.47% ; -0.12σ). This ratio depends only on the GTE orbit indices b_s , b_c , and the generation/family structure, with no CP-phase dependence (**A**).

Cross-program precision ranking (A). Among all GTE arithmetic predictions with PDG comparisons, $\lambda = 9/40$ achieves the smallest relative error. The three most precise single-formula GTE predictions are:

Table 7: Most precise single-formula GTE quantitative predictions. The λ prediction (this paper) is Lean-certified at $\mathbf{A}_{\text{Lean}}$. The $\sin^2\theta_W = 3/13$ prediction appears in [7] (P31, $\mathbf{A}_{\text{Lean}}$ cond.). The bare-coupling value $\sin^2\theta_W^{\text{bare}} = 3456/15101$ is Lean-certified in P01 and P27 from the exact gauge coupling rationals $g_1^2 = 16/125$, $g_2^2 = 2329/5400$ ($\mathbf{A}_{\text{Lean}}$).

Prediction	GTE formula	GTE value	Status	% error
λ (this paper)	$N_{\text{gen}}^2/(2^{N_{\text{gen}}} N_{\text{fam}})$	$9/40 = 0.22500$	$\mathbf{A}_{\text{Lean}}$	0.000%
$\sin^2\theta_W$ [7]	N_{gen}/c_H	$3/13 \approx 0.23077$	$\mathbf{A}_{\text{Lean}}$ cond.	-0.195%
$\sin^2\theta_W^{\text{bare}}$ (P01, P27)	$g_1^2/(g_1^2 + g_2^2)$	$3456/15101 \approx 0.2289$	$\mathbf{A}_{\text{Lean}}$	-1.02%

All three predictions lie within 2% of PDG without any free parameters. The gap from $\sin^2\theta_W^{\text{bare}} = 3456/15101$ to $\sin^2\theta_W = 3/13$ is the GTE tree-level running correction $\Delta \sin^2\theta_W = 375/196313 \approx 0.00191$ ($\mathbf{A}_{\text{Lean}}$, structured as a rational combination of N_{gen} , N_{fam} , c_H).

9 Discussion

Connection to P28: from Rule 110 to CKM arithmetic. The starting point of the derivation chain in this paper is the machine-certified result of [1] (P28) that the SM generation orbit is uniquely forced to be a Rule 110 cellular automaton orbit (CUP-4, $\mathbf{A}_{\text{Lean}}$). Rule 110 was identified in Wolfram’s survey of elementary CA rules [10] and in Wolfram’s broader program of computational models for fundamental physics [11] as the unique binary rule with a published proof of computational universality [8, 9]. P28 gives this identification a structural derivation: Rule 110 is not just any computationally universal CA but the *unique* rule consistent with the SM winding structure and vacuum transparency. The MDL-minimal $\mathbb{Z}_7$ extension f_{MDL} is then uniquely fixed (CUP-12), and all of the arithmetic in the present paper — the quark orbit indices b_u, b_d, b_c, b_s, b_b , the Wolfenstein parameter $\lambda = N_{\text{gen}}^2/(2^{N_{\text{gen}}} N_{\text{fam}})$, and the cross-sector identity $R_b = 3/8$ — are structural consequences of the CA orbit that P28 forces. The GTE framework thus derives the specific CKM arithmetic from the Rule 110 orbit structure, rather than treating Rule 110 as a given.

Connection to P31: the cross-sector identity $R_b = \sin^2\theta_W(M_{\text{GUT}})$. The companion paper [7] (P31) derives the electroweak mixing angle $\sin^2\theta_W = 3/13$ from the palindrome decomposition of the f_{MDL} neighborhoods, and the GUT-scale value $\sin^2\theta_W(M_{\text{GUT}}) = 3/8$ as an unconditional $\mathbf{A}_{\text{Lean}}$ result. The central structural result of the present paper, Theorem 6.1, states that the CKM unitarity triangle radius $R_b = 3/8$ equals this same GUT-scale Weinberg angle. Both equalities hold because they are two independent projections of the single ratio $N_{\text{gen}}/2^{N_{\text{gen}}} = 3/8$ — the generation-filling fraction of the GUT orbit — onto the electroweak and flavor sectors respectively.

This bridge has no explanation within the Standard Model: the fact that $R_b = \sin^2\theta_W(M_{\text{GUT}})$ is an arithmetically forced identity in GTE, not a numerical coincidence. Its machine certification (`ckm_unitarity_triangle_radius_eq_gut_weinberg`, $\mathbf{A}_{\text{Lean}}$, zero sorry) rests on the same Lean theorem `ngen_plus_nfam_eq_pow2` that underpins the P31 GUT formula.

Open problems. Three open problems were identified; two have been resolved. One open problem remains for future work.

(1) *Derive all quark N_{eff} formulas from GTE axioms. **Resolved.*** The quark cascade indices are GTE-derived at $\mathbf{A}_{\text{Lean}}$ (machine-certified). The three-step derivation chain — (i) MDL doublet-pairing ($\mathbf{A}/\mathbf{D}$, narrows to 12 candidates), (ii) Mersenne cascade discriminator ($\mathbf{A}_{\text{Lean}}$, §26–§30, narrows to 2), (iii) $\text{SU}(2)_L$ charge assignment ($\mathbf{A}/\mathbf{D}$, Z_7 -alignment discriminates canonical) — uniquely selects $N_{\text{eff}}(u) = N_{\text{gen}}^2 = 9$ and $N_{\text{eff}}(d) = N_{\text{fam}} = 5$. The joint Lean 4 certification theorem `six_quark_neff_complete` (`GUTStructure.lean` §34, commit c31b26c, zero sorry) packages all six values in a single 6-conjunct statement. A formal derivation of step (i) from GTE cascade axioms remains $\mathbf{A}/\mathbf{D}$ and is a target for future work.

(2) *Derive the $\lambda^3/(2c_H)$ threshold correction from first principles.* The residual $\sin^2\theta_W^{\text{PDG}} - 3/13 \approx 0.000441$ is matched to 0.7% by $\lambda^3/(2c_H) \approx 0.000438$, identified as the $k = N_{\text{gen}}$ orbit-average threshold correction ($\mathbf{A}_{\text{Lean}}$, established in [7] §7). A first-principles derivation of this formula from the $\mathbb{Z}_5$ orbit structure — showing that the k -th orbit-average correction scales as $\lambda^k/(2c_H)$ with $k = 1$ giving the leading Cabibbo suppression and $k = N_{\text{gen}} = 3$ giving the EW threshold — would provide a physical first-principles basis for this $\mathbf{A}_{\text{Lean}}$ arithmetic result and unify the CKM and EW threshold structures within a single orbit-average framework.

(3) *Prove the top quark formula from first principles. **Resolved (arithmetic; physical identification pending).*** The top quark cascade index $b_t = 2^{c_W} \cdot N_{\text{gen}} \cdot N_{\text{fam}} \cdot c_W = 337920$ is machine-certified at $\mathbf{A}_{\text{Lean}}$: `neff_t_formula` (`GUTStructure.lean` §20, commit c4b0ae5), `neff_t_from_even_step` (`GUTStructure.lean` §25, norm_num, commit a0f7750), and `bt_from_cascade_quotient` (`GUTStructure.lean` §25, commit a0f7750); zero sorry. The theorem `neff_t_from_even_step` certifies the factored form $2^{c_W} \cdot N_{\text{gen}} \cdot N_{\text{fam}} \cdot c_W = 337920$ via explicit let-bindings, making the structural role of each factor ($c_W = c_H - 2 = 11$ from the W-boson cascade depth) arithmetically transparent. Moreover, the W-boson cascade depth $c_W = 11 = \lfloor 823/73 \rfloor$ emerges as the first GTE cascade quotient at the lepton seed, independently of injected parameters (`q_11_eq_c_w`, $\mathbf{A}_{\text{Lean}}$, zero sorry). The mass ratio $b_t/b_b = 337920/8191 \approx 41.26$ matches the PDG value $m_t/m_b \approx 41.46$ to 0.49%. The formal derivation of c_W from GTE cascade axioms remains $\mathbf{A}/\mathbf{D}$ and is a target for future work.

(4) *Derive the $U(N_{\text{gen}})$ degree-of-freedom identification from GTE orbit structure.* The physical identification of $N_{\text{gen}}^2 = 9$ as the real dimension of $U(N_{\text{gen}})$ — and of $2^{N_{\text{gen}}} \times N_{\text{fam}} = 40$ as the GUT-orbit capacity — is $\mathbf{A}/\mathbf{D}$ (§4). Three parallel interpretations exist: (i) information density, (ii) cross-sector projection $\lambda = \sin^2\theta_W(M_{\text{GUT}}) \cdot N_{\text{gen}}/N_{\text{fam}}$, and (iii) probability flow (each covered at $\mathbf{A}/\mathbf{D}$ in §4). A full $\mathbf{A}$ derivation requires a formal GTE orbit structure theorem identifying the real dimension of the generation-mixing group with N_{gen}^2 .

(5) *Upgrade the Jarlskog invariant to $\mathbf{A}_{\text{Lean}}$.* The GTE prediction $J_{\text{GTE}} = \lambda^6 A^2 \bar{\eta} = 2.999 \times 10^{-5}$ sits 1.81σ below PDG, an offset arising from the multiplicative accumulation of small individual errors in λ , A , $\bar{\rho}$, and $\bar{\eta}$. Each factor is individually at $\mathbf{A}_{\text{Lean}}$; the product inherits the accumulated uncertainty. The exact rational expression $J_{\text{GTE}} = \lambda^6 A^2 \bar{\eta}^2 (1 - \lambda^2/2)^2$ is machine-certified as a positive rational number (`jarlskog_invariant_gte_formula`, §72, $\mathbf{A}_{\text{Lean}}$, zero sorry). Full $\mathbf{A}_{\text{Lean}}$ upgrade of the numerical prediction requires Lean formalization of the quark cascade N_{eff} derivation (step (i) above) — a long-term Lean target.

Joint Selection Theorem and Double Mersenne Uniqueness. The arithmetic coincidence that both $N_{\text{fam}} = 5$ and $c_H = 13$ are Mersenne prime exponents is not a numerical accident. The Joint Selection Theorem (`joint_selection_theorem`, `GUTStructure.lean` §21, commit 8180fca, `A_Lean`, zero sorry) certifies that among all primes $p \leq 23$, $N_{\text{fam}} = 5$ is the unique prime satisfying simultaneously (i) $\mathbb{Z}_5$ ring transitivity under Rule 110, and (ii) a Mersenne prime Higgs capacity $c_H = N_{\text{gen}} + 2p$. The double Mersenne exponent identity (`double_mersenne_exponent_identity`, `A_Lean`): (a) $2^{N_{\text{fam}}} - 1 = 31$ is prime ($N_{\text{fam}} = 5$ is a Mersenne exponent); (b) $2^{c_H} - 1 = 8191$ is prime ($c_H = 13$ is a Mersenne exponent); (c) $c_H - 2N_{\text{fam}} = N_{\text{gen}}$ (cascade pivot $p_5 - 2p_3 = 13 - 10 = 3$); (d) $c_H = N_{\text{gen}} + 2N_{\text{fam}}$ (cascade formula). The GTE triple $(3, 5, 13)$ is the unique triple surviving both filters.

Arithmetic origin vs. dynamical EWSB. The Wolfenstein parameters derived in this paper — $\lambda = 9/40$, A , R_b , $\bar{\eta}$ — arise from $\mathbb{Z}_7$ orbit arithmetic via the Algebraic Descent Theorem (`A_Lean`, P41 [12]). They determine *what* the CKM mixing values are; they do not address the dynamical mechanism by which quark mass eigenstates mix. The dynamical EWSB mechanism that sets the absolute Fermi scale $G_F \sim v_H^{-2}$ is established separately via the SRRG fixed-point derivation (P27 [13], `A/D`), and the $\text{SU}(2)_L$ gauging of the weak current is an open problem at CatD (*cf.* P35 [14] §open problems). These are complementary results; the orbit arithmetic provides the structural ratios, and the SRRG + $\text{SU}(2)_L$ results provide the dynamical substrate.

A Lean 4 Certification Inventory

All theorems below are in the `ugp-lean` repository, zero bare sorry. Commit c4b0ae5 certifies `GUTStructure.lean` §14–§15 and §20 entries. Commit c31b26c certifies `GUTStructure.lean` §34 entries (capstone). Commit a0f7750 certifies `GUTStructure.lean` §25 entries (W-gateway, even-step formula). Commit 143100b certifies `GUTStructure.lean` §30 entries (Mersenne cascade discriminator). Commit 9ba0146 certifies `GUTStructure.lean` §33 entries ($\text{SU}(2)_L$ charge assignment). Commit ca5f831 certifies §20 CP irrationality chain theorems (`neff_s_not_prime`, `neff_b_s_coprime`, `cp_violation_irrationality_chain`, `neff_t_factors`, `top_bottom_ratio_q`). Commit 8180fca certifies §21 Joint Selection Theorem and double Mersenne identities.

Theorem	Module	Tactic	Commit	0-sorry
wolfenstein_lambda_formula	GUTStructure §14	norm_num	c4b0ae5	✓
wolfenstein_lambda_value	GUTStructure §14	norm_num	c4b0ae5	✓
ckm_dof_count	GUTStructure §14	norm_num	c4b0ae5	✓
wolfenstein_A_sq_rational	GUTStructure §15	norm_num	c4b0ae5	✓
ckm_unitarity_triangle_radius_eq_gut_weinberg	GUTStructure §15	alias	c4b0ae5	✓
ckm_from_gte_arithmetic	GUTStructure §15	norm_num	c4b0ae5	✓
neff_u_eq_ngen_sq	GUTStructure §15	norm_num	c4b0ae5	✓

Theorem	Module	Tactic	Commit	0-sorry
neff_d_eq_nfam	GUTStructure §15	norm_num	c4b0ae5	✓
neff_c_eq_nfam_poly	GUTStructure §15	norm_num	c4b0ae5	✓
neff_s_eq_gen_higgs_form	GUTStructure §15	norm_num	c4b0ae5	✓
neff_b_eq_mersenne	GUTStructure §15	norm_num	c4b0ae5	✓
neff_b_is_prime	GUTStructure §15	norm_num	c4b0ae5	✓
chiggs_is_5th_mersenne_exp	GUTStructure §15	nn+nd	c4b0ae5	✓
bb_bs_product_not_square	GUTStructure §15	bounded arith.	c4b0ae5	✓
ngen_plus_nfam_eq_pow2	GUTStructure §2	norm_num	c4b0ae5	✓
gte_family_capacity_identity	GUTStructure §10	alias	c4b0ae5	✓
gut_weinberg_angle_pow2	GUTStructure §3	norm_num	c4b0ae5	✓
neff_t_formula	GUTStructure §20	norm_num	c4b0ae5	✓
neff_t_factors	GUTStructure §20	norm_num	ca5f831	✓
top_bottom_ratio_q	GUTStructure §20	norm_num	ca5f831	✓
neff_t_from_even_step	GUTStructure §25	norm_num	a0f7750	✓
q_l1_eq_c_w	GUTStructure §25	norm_num	a0f7750	✓
bt_from_cascade_quotient	GUTStructure §25	norm_num	a0f7750	✓
charm_triple_as_mersenne	GUTStructure §25	norm_num	a0f7750	✓
bb_mersenne_bt_not	GUTStructure §30	exact	143100b	✓
cascade_c_pair_mersenne_unique	GUTStructure §30	exact	143100b	✓
quark_cascade_mersenne_discriminator	GUTStructure §30	exact	143100b	✓
su2l_charge_assignment_z7_discriminator	GUTStructure §33	exact	9ba0146	✓

Theorem	Module	Tactic	Commit	0-sorry
six_quark_neff_complete	GUTStructure §34	norm_num	c31b26c	✓
quark_g1_canonical_assignment	GUTStructure §34	norm_num	c31b26c	✓
quark_neff_all_distinct	GUTStructure §34	norm_num	c31b26c	✓
neff_s_not_prime	GUTStructure §20	nd	ca5f831	✓
neff_b_s_coprime	GUTStructure §20	nd	ca5f831	✓
tan_gamma_numerator_not_square	GUTStructure §20	nn+ linarith	ca5f831	✓
cp_violation_irrationality_chain	GUTStructure §20	exact	ca5f831	✓
bb_bs_sqrt_floor	GUTStructure §20	nn	ca5f831	✓
joint_selection_theorem	GUTStructure §21	nn+decide	8180fca	✓
double_mersenne_exponent_identity	GUTStructure §21	nn	8180fca	✓
cH_eq_mersenne_exp_nfam	GUTStructure §40	nd	—	✓
cH_structural_triple_identity	GUTStructure §40	nn	—	✓
cH_palindrome_formula	GUTStructure §40	nn	—	✓
mersenne_exp_first_five	GUTStructure §40	nd	—	✓
quark_g1_permutation_rule_lex_derivation	GUTStructure §41	norm_num	6010180	✓
quark_g1_lex_min_certificate	GUTStructure §41	norm_num	6010180	✓
wolfenstein_A_satisfies_eq	GUTStructure §72	norm_num	—	✓
wolfenstein_A_bounds	GUTStructure §72	norm_num	—	✓
cabibbo_angle_exists	GUTStructure §72	exact	—	✓

Theorem	Module	Tactic	Commit	0-sorry
<code>gamma_cp_tan_value</code>	GUTStructure §72	<code>norm_num</code>	—	✓
<code>gamma_cp_bounds_deg</code>	GUTStructure §72	<code>norm_num</code>	—	✓
<code>rho_bar_eta_bar_bounds</code>	GUTStructure §72	<code>norm_num</code>	—	✓
<code>rho_bar_sq_exact</code>	GUTStructure §72	<code>norm_num</code>	—	✓
<code>eta_bar_sq_exact</code>	GUTStructure §72	<code>norm_num</code>	—	✓

Table 8: Complete Lean 4 certification inventory. “nn” = `norm_num`; “nd” = `native_decide`; “—” = commit pending on `ugp-lean` main branch. All rows are $\mathbf{A}_{\text{Lean}}$ (machine-certified, zero `sorry`). The §34 capstone theorem `six_quark_neff_complete` packages all six quark N_{eff} values in a single 6-conjunct statement (commit `c31b26c`). Commit `ca5f831` certifies §20 CP irrationality chain and top quark factors; commit `8180fca` certifies §21 Joint Selection and double Mersenne identities; §40 entries certify the $c_H = p_{N_{\text{fam}}}$ structural identities (commit pending); §72 entries certify the Wolfenstein parameter arithmetic, Cabibbo angle existence, CP-angle and CKM-apex interval bounds, and the exact rational squared values $\bar{\rho}^2 = 7533/315680$ and $\bar{\eta}^2 = 73719/631360$ (all 8 theorems zero `sorry`; commit pending).

B Quantitative Predictions Summary

The complete quantitative predictions for all GTE CKM observables are collected in Tables 5 and 6 in §8. The structural ratio prediction $\lambda/R_b = N_{\text{gen}}/N_{\text{fam}} = 3/5$ appears in (8). The CP angle prediction $\gamma = 65.67^\circ$ appears in (14). The Jarlskog invariant is computed in §8.

C Scripts and Reproduction

All numerical results in this paper are reproducible using the scripts in the paper directory. Primary script: `scripts/ckm_from_gte.py` (all four Wolfenstein parameters, full $O(\lambda^4)$ CKM matrix, Jarlskog invariant). Supporting scripts: `scripts/cp_violation_mersenne.py` (Mersenne irrationality verification and CP angle computation) and `scripts/wolfenstein_parameter.py` ($\lambda = 9/40$ arithmetic verification). See `REPRODUCE.md` in the paper directory for reproduction instructions.

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